

**FELIX ATUHAIRE**

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[fatuhaire@must.ac.ug](mailto:fatuhaire@must.ac.ug) ([atuhairef53@gmail.com](mailto:atuhairef53@gmail.com)) ([fxa351@case.edu](mailto:fxa351@case.edu))**RESEARCH INTERESTS**

My research lies at the intersection of engineering, computer science, and medicine. My research areas are brain-computer interfaces and signal processing.

**EDUCATION**

|                                           |                                                     |             |
|-------------------------------------------|-----------------------------------------------------|-------------|
| PhD Student (Biomedical Engineering)      | Case Western Reserve University, Ohio               | In progress |
| MSc Biomedical Engineering                | University of Cape Town, Cape Town                  | 2018-2020   |
| BSc Computer & Communications Engineering | Mbarara University of Science & Technology, Mbarara | 2011-2015   |

**DISSERTATION**

*Reconstruction of three-dimensional facial geometric features related to fetal alcohol syndrome using adult surrogates (2019-2020)*

This dissertation investigated techniques for reconstructing 3D human faces from single 2D images of the face based on a 3D morphable model (3DMM) of the face. The 3DMM was built from 3D surface scans of the human face obtained using a scanning device. We applied probabilistic methods to estimate 3D face reconstructions from single 2D facial images using the 3DMM as prior knowledge.

*Adviser:* Ass. Prof. Tinashe Mutsvangwa

*Committee members:* Prof. Thomas Franz & Dr. Frances Robertson

*Outside reader:* Dr. Michael Suttie

*Design and development of a solid waste monitoring system (2014-2015)*

This final year project developed a tool for monitoring solid waste levels in municipal and city bins in Mbarara city. We integrated both hardware and software tools to achieve the project's objectives. The project aimed to improve sanitation in the city by sending timely reminders to the waste collection office and generating reports for the city managers.

*Supervisor:* Dr. Simon Kawuma

**RESEARCH EXPERIENCE**

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| <b>Graduate Student</b> , Lab of Ass. Professor Tinashe Mutsvangwa                                                                                                                                                                                                                                                                                                 | 2018-2020 |
| Medical image-based inference & distributed diagnostics (Mi2D2), University of Cape Town, Cape Town                                                                                                                                                                                                                                                                |           |
| <ul style="list-style-type: none"><li>○ Built a 3D shape and appearance model of the human face and used it to estimate a 3D face reconstruction given a single 2D facial image.</li><li>○ Discovered that this approach could be applied to analyze facial images for several disorders affecting the facial region, especially fetal alcohol syndrome.</li></ul> |           |

**TEACHING EXPERIENCE**

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|---------------|
| <b>Assistant Lecturer</b>                                                                                                                                                                                                                    | Department of Biomedical Sciences & Engineering | Mbarara University of Science & Technology | 2021- Present |
| <ul style="list-style-type: none"><li>○ Designed, instructed, and assessed courses including medical imaging, biomechanics, applied physics, computer programming, and introduction to machine learning at an undergraduate level.</li></ul> |                                                 |                                            |               |
| <b>Teaching Assistant</b>                                                                                                                                                                                                                    | Department of Computer Engineering              | Mbarara University of Science & Technology | 2016-2017     |
| <ul style="list-style-type: none"><li>○ Assisted in tutoring and grading undergraduate students in various courses including engineering mathematics, applied physics, and object-oriented programming with Java.</li></ul>                  |                                                 |                                            |               |

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**COURSES TAUGHT**

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| ○ Medical imaging systems, Biomechanics, Engineering physics, Biosensing & Measurements, Object-oriented programming with Java, Fundamentals of machine learning | Current   |
| ○ Transducers in biomedical instrumentation, Applied physics, Microprocessors, Robotics in medicine                                                              | 2021-2022 |
| ○ Biomechanics, Engineering Mechanics, Digital Communication Systems                                                                                             | 2021-2022 |
| ○ Engineering Mathematics, Digital Signal Processing, Electrical circuits and instrumentation                                                                    | 2016-2017 |

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**SCHOLARSHIPS AND AWARDS**

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|-------------------------------------------------------------------------------------------------|------|
| International and Refugee Scholarship                                                           | 2019 |
| ○ Awarded funding for research and upkeep while studying at the University of Cape Town         |      |
| African Biomedical Engineering Mobility Scholarship                                             | 2018 |
| ○ Received funding to study a master's in biomedical engineering at the University of Cape Town |      |
| The Convocation Awards of Mbarara University of Science and Technology                          | 2016 |
| ○ Received a prize for being the best student in my class with a first-class honours degree     |      |
| The Annual Faculty of Computing & Informatics Awards                                            | 2015 |
| ○ Awarded a prize for the best undergraduate research project                                   |      |
| The Annual Faculty of Computing & Informatics Awards                                            | 2014 |
| ○ Won a prize for the first runner-up undergraduate research project                            |      |

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**CONFERENCES, WORKSHOPS, AND SEMINARS**

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- Machine learning training by faculty from Ghent University in Belgium under the University as Facilitator for Community-based Sustainable Solutions to Demographic Challenges in South-Western Uganda (UCoBS), Mbarara, Uganda, 2024
- Artificial Intelligence, Machine Learning, and Bioinformatics training under the Africa Centre of Excellence in Materials Product Development & Nanotechnology (MAPRONANO ACE), Mbarara, Uganda, 2023
- Strengthening engineering education and research (SEER) training under the Academics Without Borders / Universitaires Sans Frontieres, Canada (Virtual), 2021
- Engineering education for development at the auspices of higher education partnerships for sub-Saharan Africa, Kampala, Uganda, 2021
- Machine Learning Summer School, Stellenbosch, South Africa, 2019
- Probabilistic morphable model fitting workshop through the University of Basel and University of Cape Town, Cape Town, South Africa, 2018
- The 4<sup>th</sup> National Conference on Communications organised by Uganda Communication Commission (UCC) in conjunction with Mbarara University of Science and Technology (MUST), Mbarara, 2017
- Global Surgery Hack-a-thon organised by CAMTech Uganda, Mbarara, Uganda, 2016
- The 12<sup>th</sup> MUST annual research dissemination conference organised by the Institute of Management Sciences, Mbarara, Uganda, 2016
- The 11<sup>th</sup> MUST annual research dissemination conference with the theme “Rethinking science and technology: a case of interdisciplinary approaches to research and community development”, 2015
- The 2<sup>nd</sup> CAMTech-MUST MedTech Hack-a-thon, Mbarara, Uganda, 2014
- The 3<sup>rd</sup> national conference on communications on the theme “ICT strategic opportunities and challenges for national development”, 2014

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**SUPERVISORY & MENTORING EXPERIENCE**

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- Supervised and mentored 15 undergraduate students in four different research projects

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## ACADEMIC SERVICE

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Faculty Undergraduate Laboratory Coordinator                                                                                                                                                                                                                                          | 2021-   |
| ○ Responsible for designing lab schedules, developing lab manuals and sharing them with staff, organising periodic meetings between lab technicians and teaching staff, preparing reports on the status of the labs, and orienting staff and students on the facility's capabilities. | Present |
|                                                                                                                                                                                                                                                                                       |         |
| Departmental Undergraduate Research Coordinator                                                                                                                                                                                                                                       | 2022-   |
| ○ Design project timelines, assign supervisors to student teams, monitor student project progress through periodic presentations, design evaluation forms, organize workshops to guide students at each stage of the project development, and assess students' project reports.       | Present |

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## TECHNICAL SKILLS

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- Programming skills in Java, Python, Scala, R, and MATLAB

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## PRESENTATIONS

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*3D reconstruction of the human face from a single 2D image towards medical image analysis*, The 8<sup>th</sup> Face Science Symposium Virtual Edition, South Africa, 2020

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## PUBLICATIONS

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[1]. **Atuhaire, F.**, Egger, B., & Mutsvangwa, T. (2023). “*Evaluating 3D human face reconstruction from a frontal 2D image, focusing on facial regions associated with fetal alcohol syndrome*”. South African Journal of Science, 119(3/4). doi:10.17159/sajs.2023/12064

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## REFEREE

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Ass. Prof. Tinashe Mutsvangwa (*Masters Mentor & Advisor*)  
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